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(54) **DISPLAY APPARATUS**

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H01L 25/075 (2006.01)
H05K 5/30 (2025.01)

(52) **U.S. Cl.**
CPC **H05K 5/0217** (2013.01); **H01L 25/0753** (2013.01); **H05K 5/30** (2025.01)

(58) **Field of Classification Search**

CPC H05K 5/0017; H05K 5/0217

USPC 361/807

See application file for complete search history.

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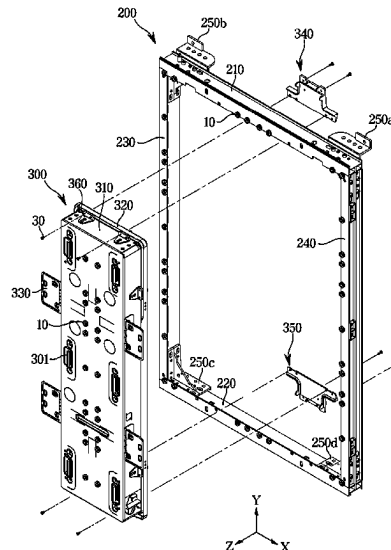
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(57) **ABSTRACT**

A display apparatus includes: a display module including a substrate on which a plurality of light emitting diodes (LEDs) are mounted; a cabinet provided to support the display module; and a circuit case attached to the cabinet. The display module is provided to be detachable from and mountable on a front side of the cabinet, and the display module is provided to be detachable from and mountable on a rear side of the cabinet.

12 Claims, 13 Drawing Sheets



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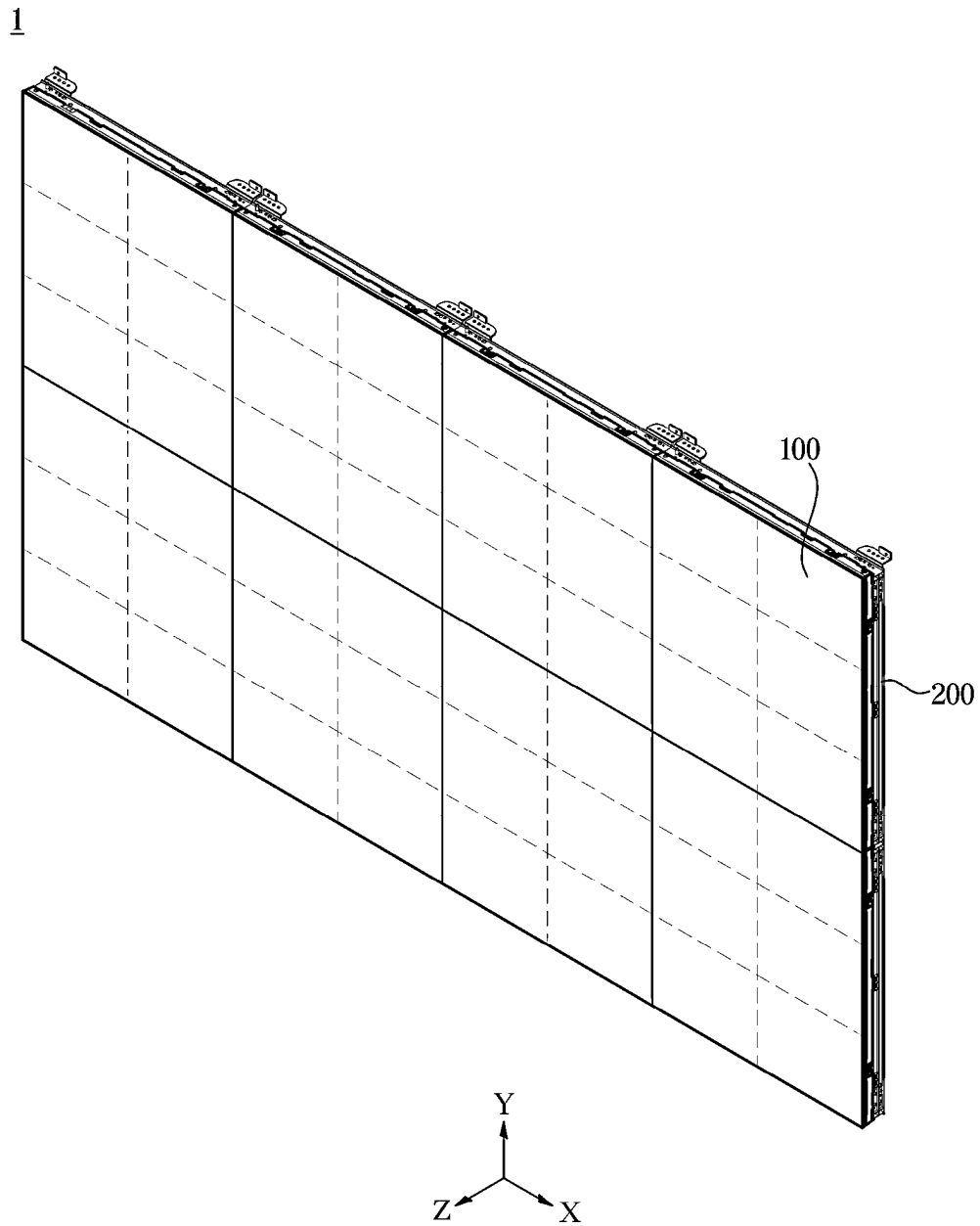
FIG. 1

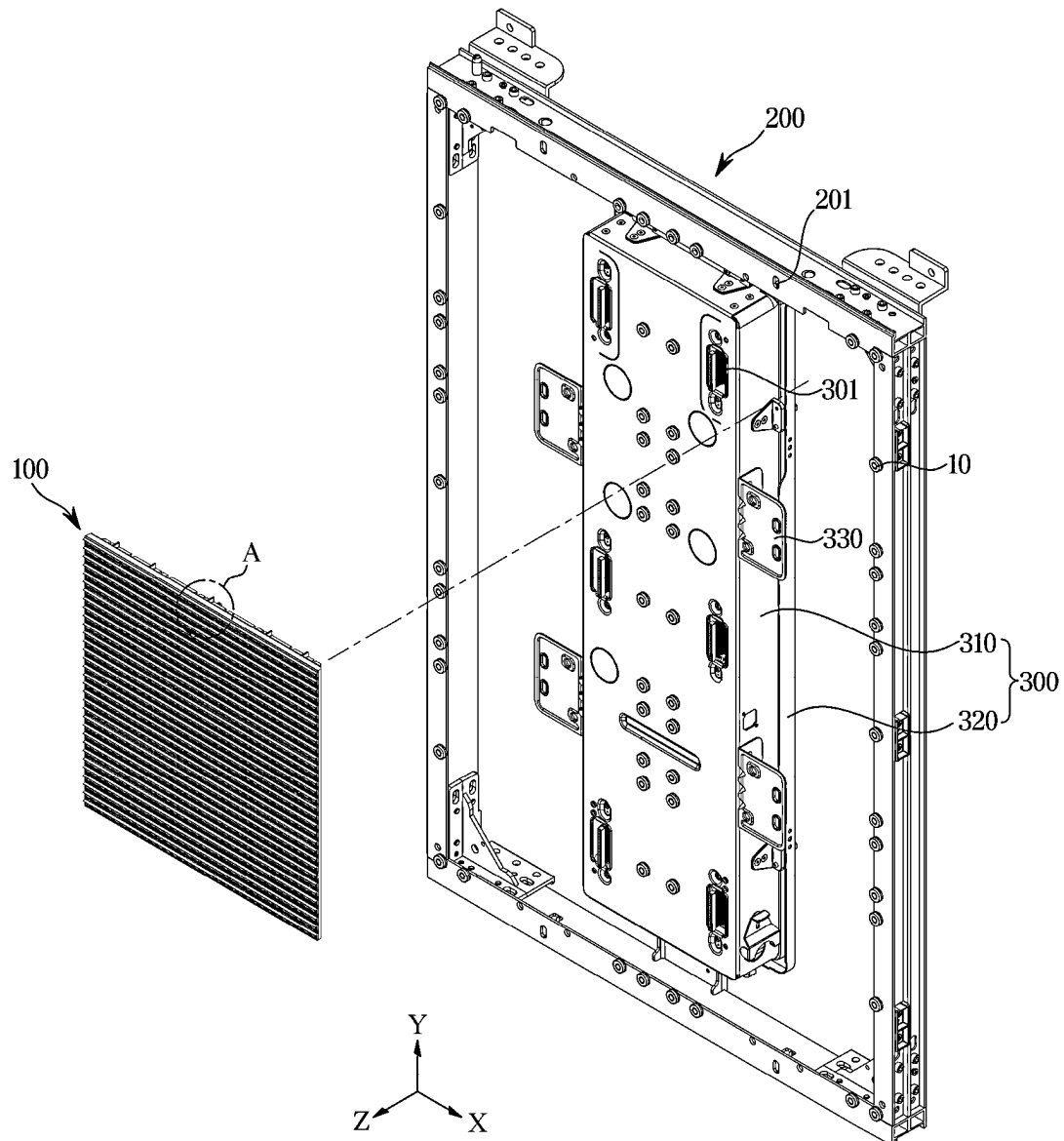
FIG. 2

FIG. 3

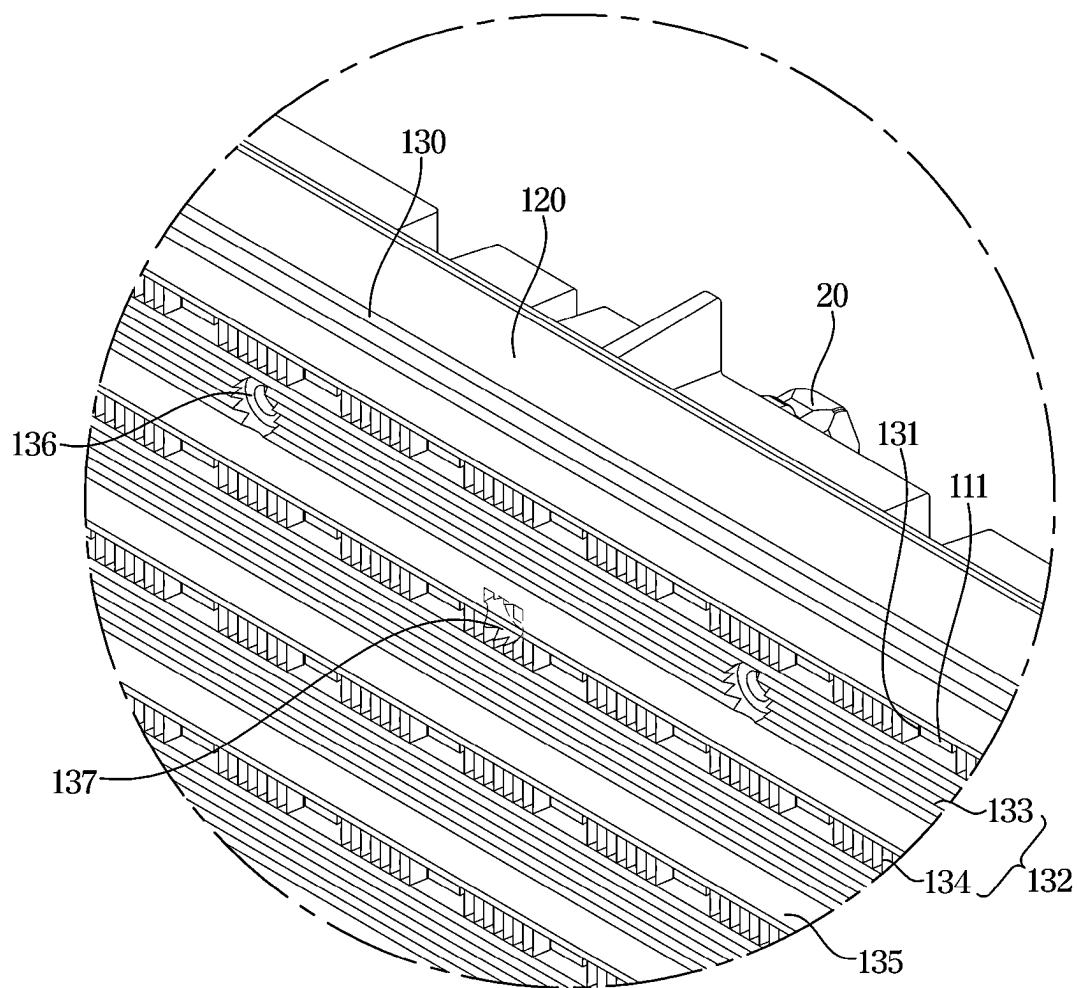


FIG. 4

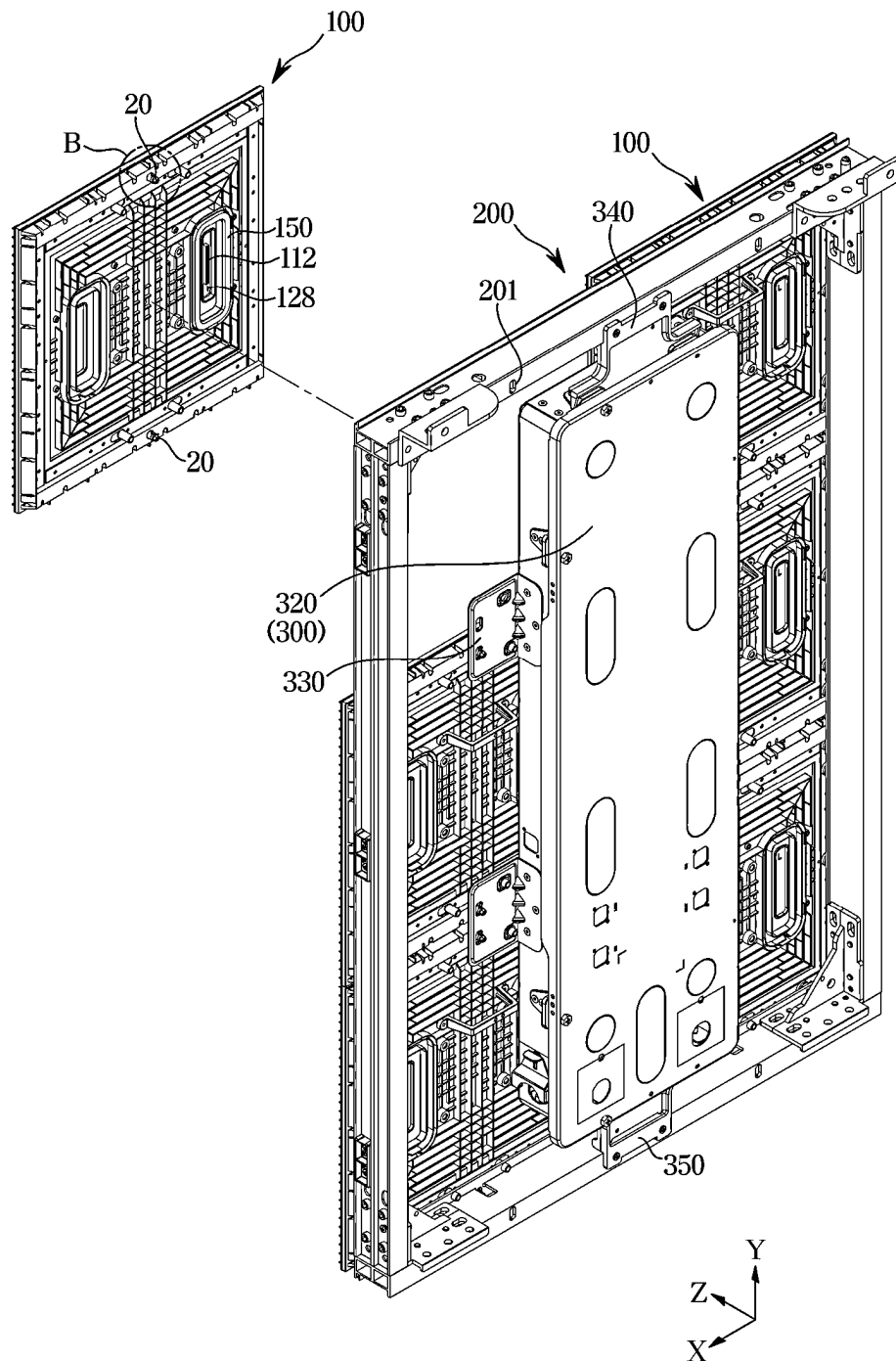


FIG. 5

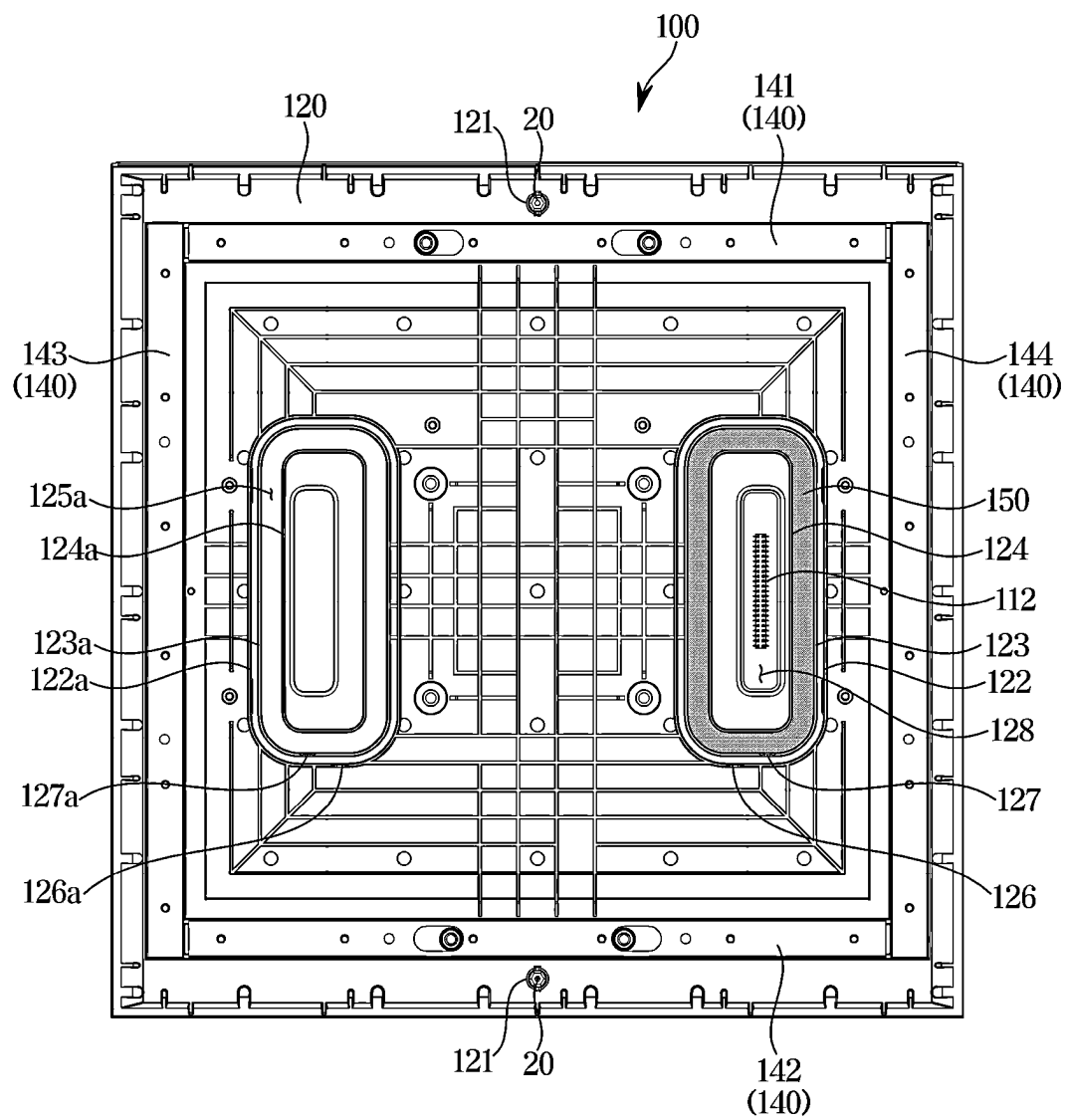


FIG. 6

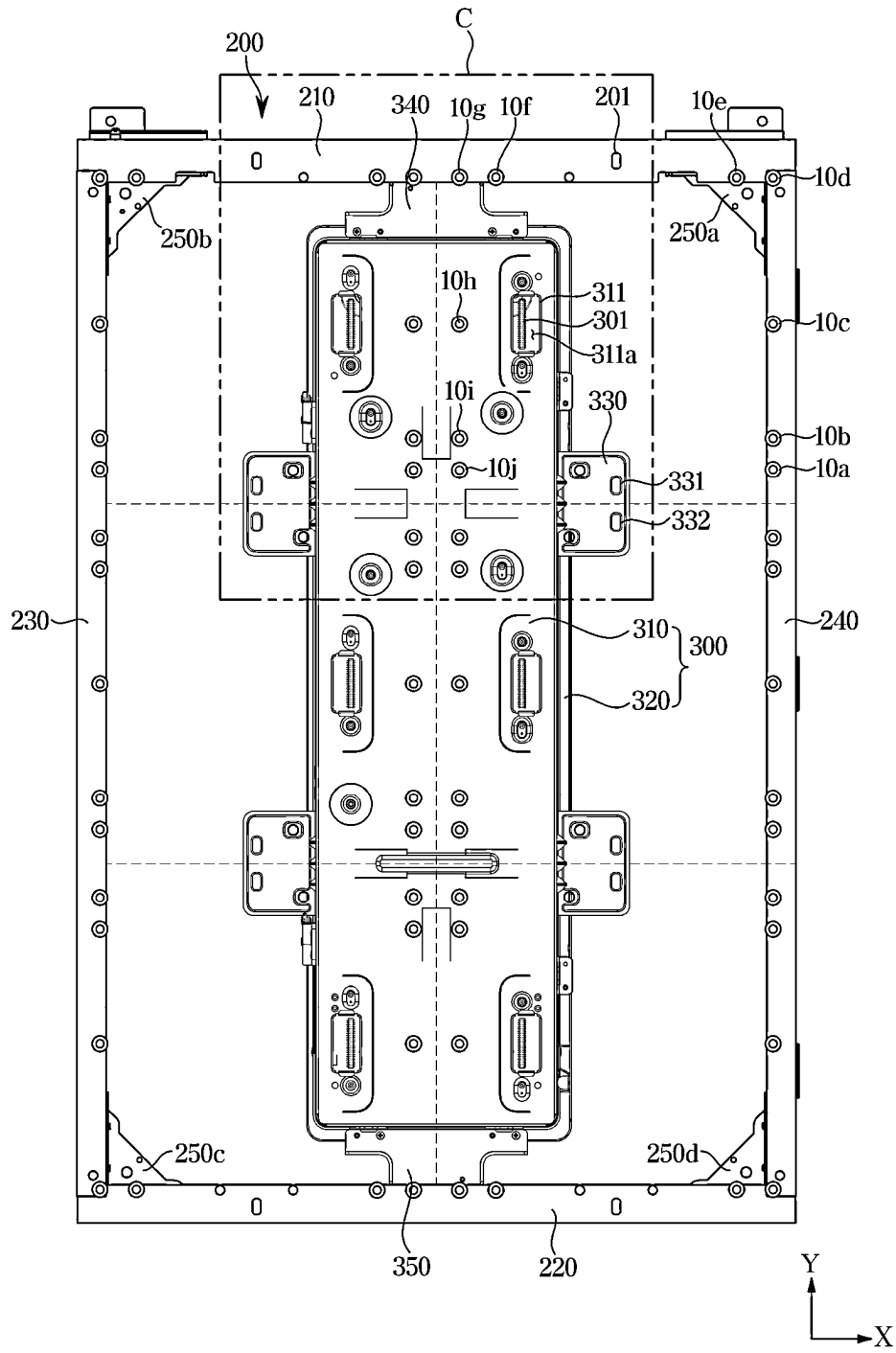


FIG. 7

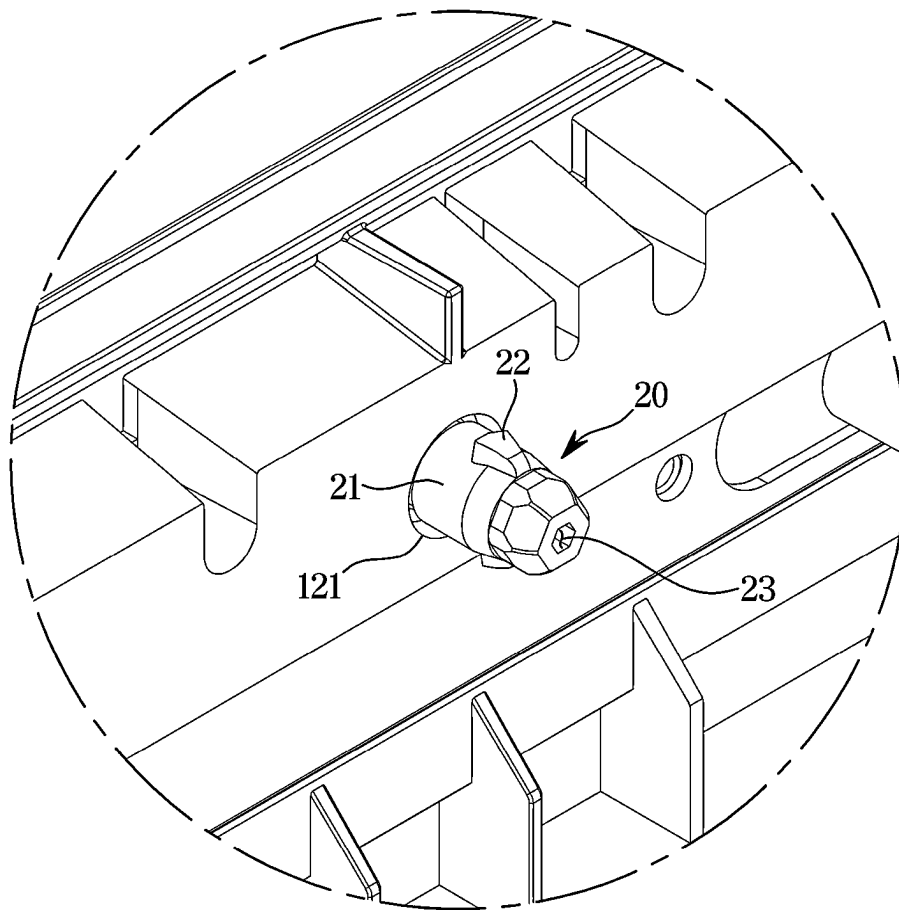


FIG. 8

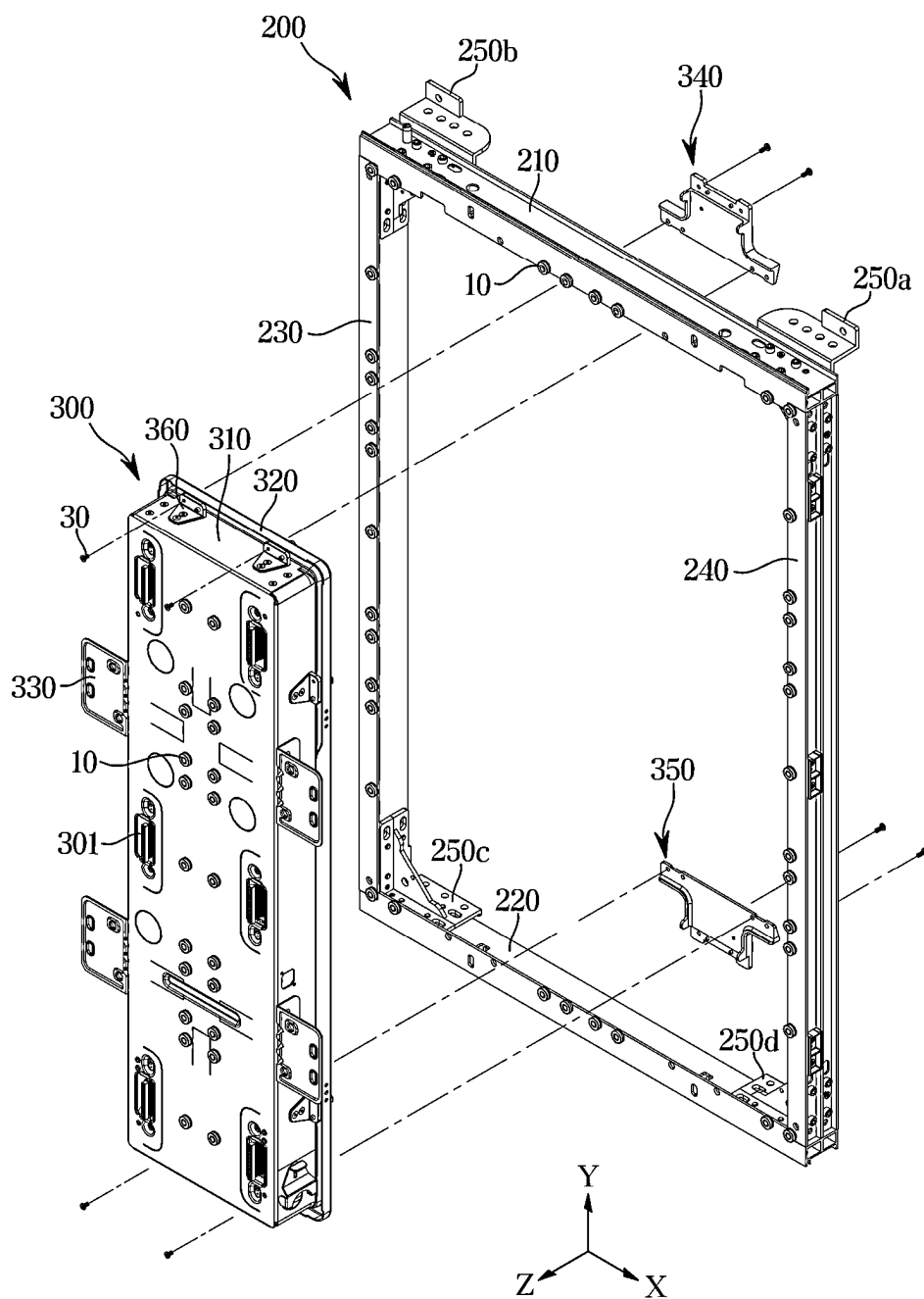


FIG. 9

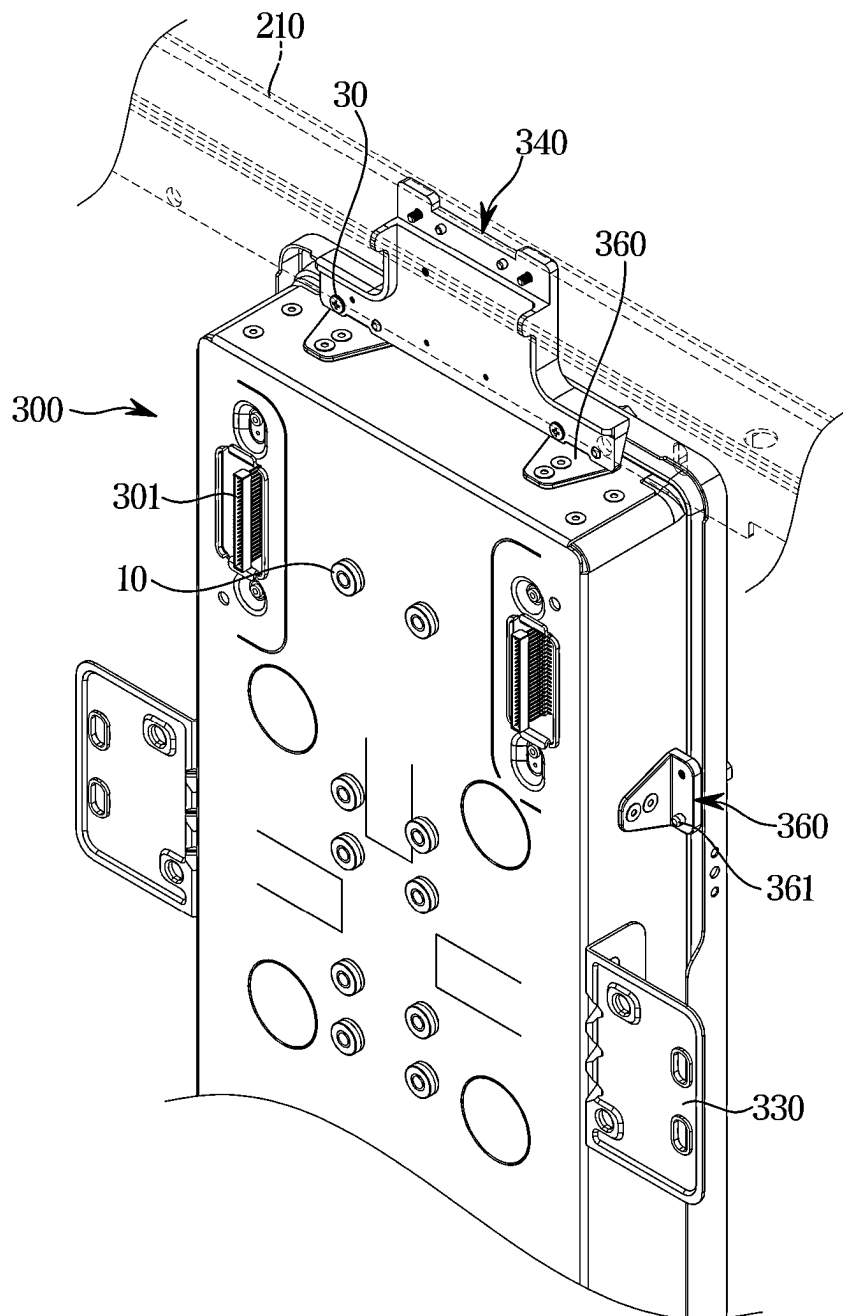


FIG. 10

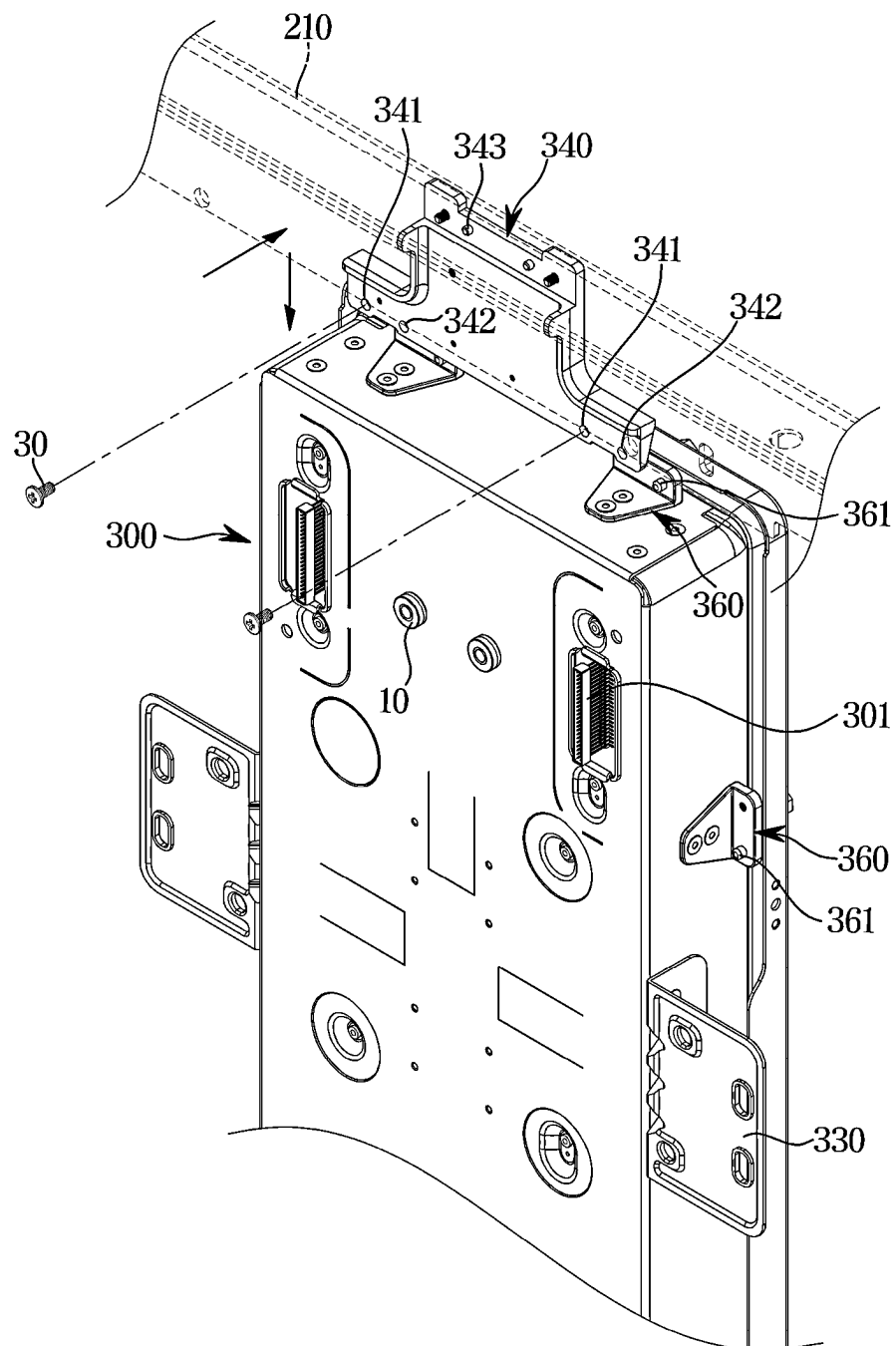


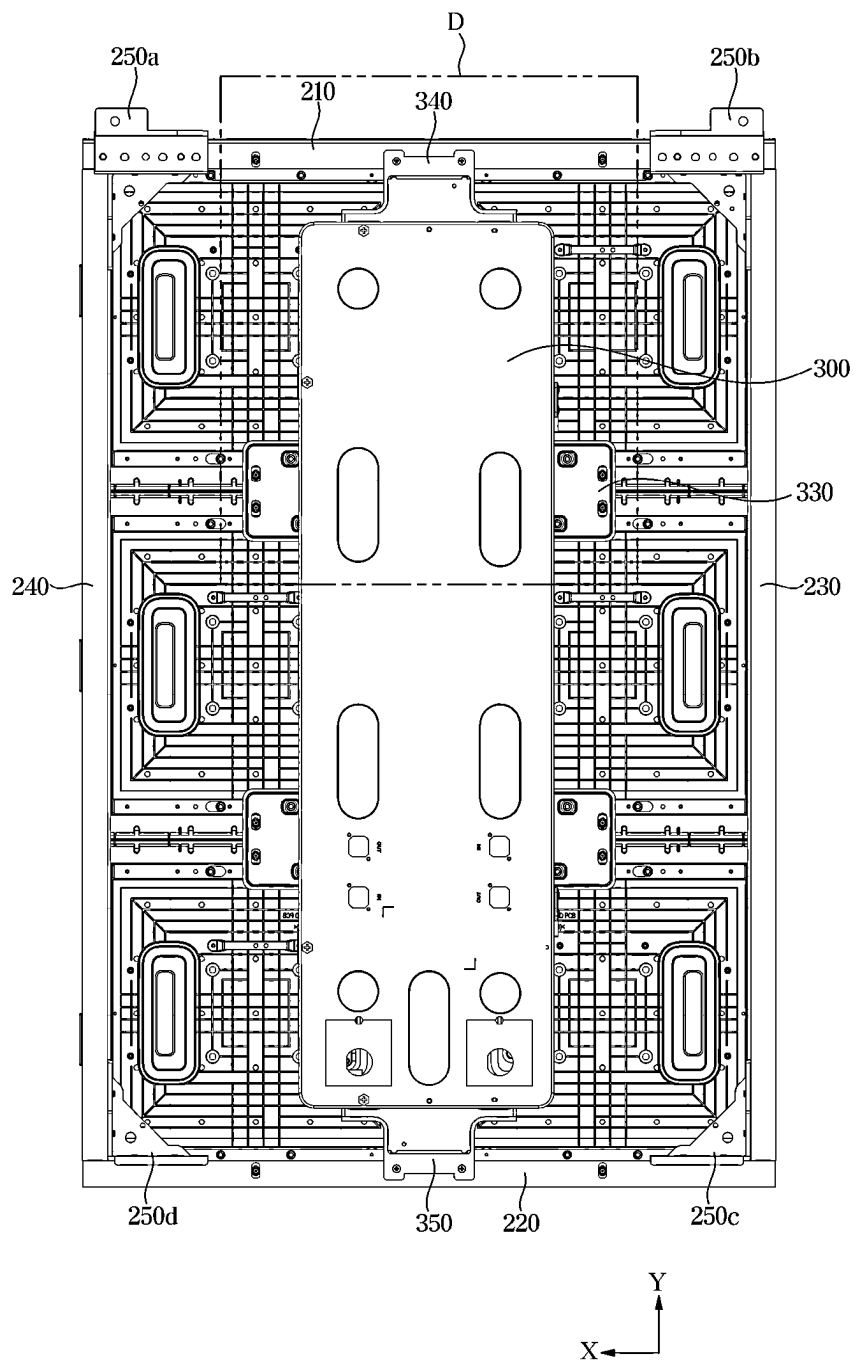
FIG. 11

FIG. 12

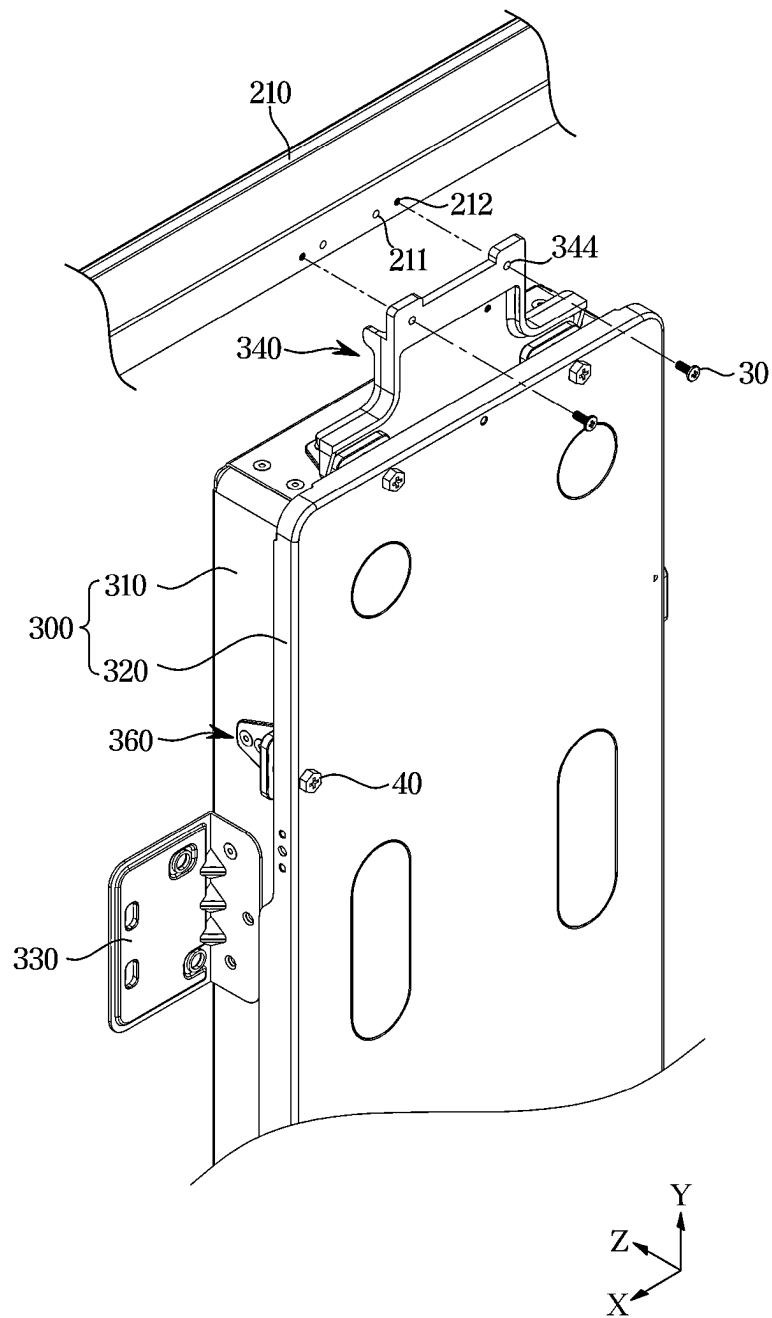
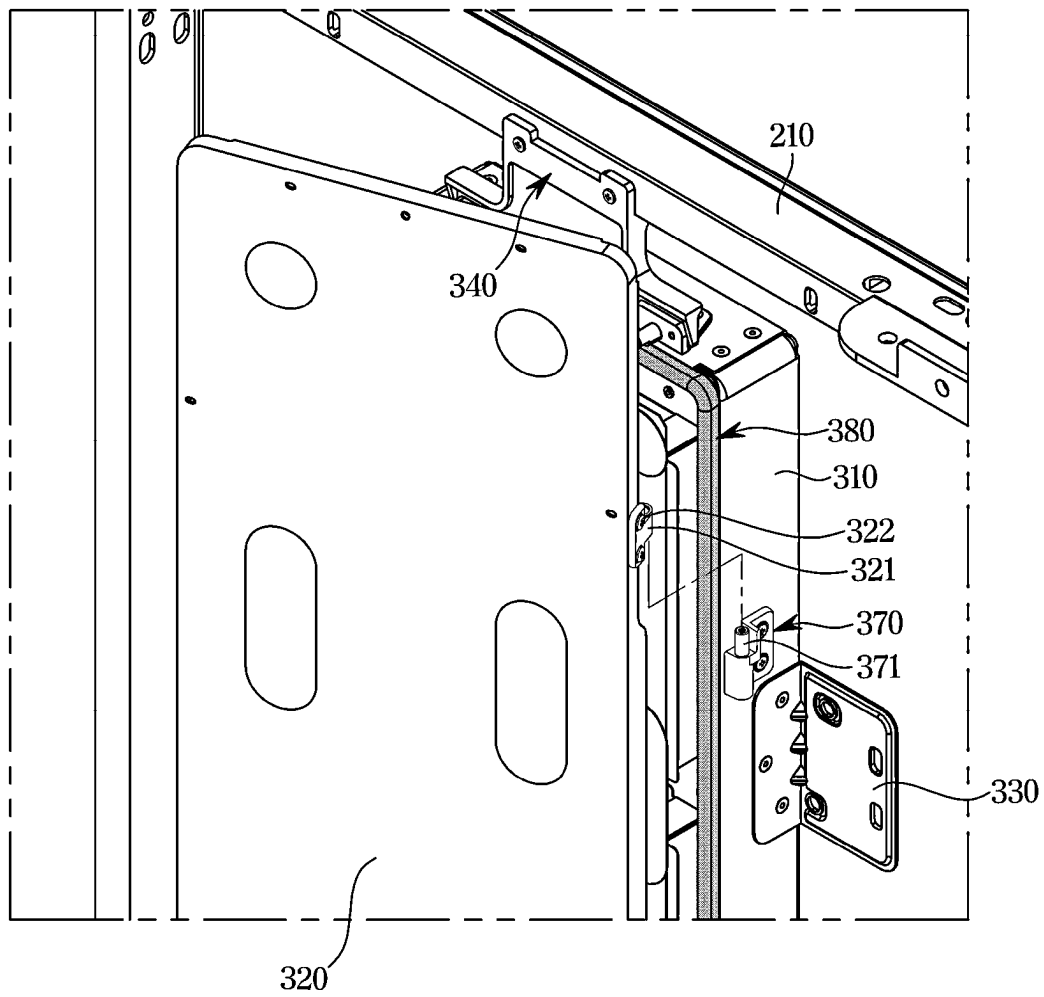


FIG. 13

DISPLAY APPARATUS**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a by-pass continuation application of International Application No. PCT/KR2023/005195, filed on Apr. 18, 2023, which is based on and claims priority to Korean Patent Application Nos. 10-2022-0089886, filed on Jul. 20, 2022, and 10-2022-0111740, filed on Sep. 2, 2022, in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein their entireties.

BACKGROUND**1. Field**

The disclosure relates to a display apparatus including a plurality of display modules and a cabinet supporting a plurality of display modules.

2. Description of Related Art

A display apparatus converts acquired or stored electrical information into visual information and displays the converted information to a user.

In order to implement a large-sized screen, the display apparatus may include a plurality of display modules and a cabinet supporting the plurality of display modules that are consecutively tiled in an upper to lower direction and a left to right direction. The cabinet supporting the plurality of display modules may be provided in plural, and the plurality of cabinets may also be provided to be consecutively tiled in an upper to lower direction and a left to right direction similar to the plurality of display modules. The plurality of display modules supported by the plurality of cabinets may form a single large screen.

A circuit case of a display apparatus may need to be detachable from a cabinet, thus the circuit case may need to be accessed from the front and rear of the display apparatus.

SUMMARY

One aspect of the disclosure provides a display apparatus in which a circuit case is detachable from a cabinet by accessing the circuit case from the front and rear of the display apparatus.

According to an aspect of the disclosure, a display apparatus includes: a display module including a substrate on which a plurality of light emitting diodes (LEDs) are mounted; a cabinet provided to support the display module; and a circuit case attached to the cabinet. The circuit case is provided to be detachable from and mountable on a front side of the cabinet, and the circuit case is provided to be detachable from and mountable on a rear side of the cabinet.

According to an aspect of the disclosure, a display apparatus in which a circuit case is detachable from a cabinet by accessing the circuit case from the front and rear of the display apparatus can be provided.

BRIEF DESCRIPTION OF THE DRAWINGS

The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

FIG. 1 is a view illustrating a display apparatus according to an embodiment;

FIG. 2 is a view illustrating a cabinet and a display module separated from the cabinet in a display apparatus according to an embodiment, when viewed from the front;

FIG. 3 is an enlarged view of portion A of FIG. 2 according to an embodiment;

FIG. 4 is a view illustrating a cabinet and a display module separated from the cabinet in a display apparatus according to an embodiment, when viewed from the rear;

FIG. 5 is a rear view illustrating a display module in a display apparatus according to an embodiment;

FIG. 6 is a front view illustrating a cabinet in a display apparatus according to an embodiment;

FIG. 7 is an enlarged view of portion B of FIG. 4 according to an embodiment;

FIG. 8 is a view illustrating a cabinet and a circuit case separated from the cabinet in a display apparatus according to an embodiment;

FIG. 9 is an enlarged view of portion C of FIG. 6 when viewed at a different angle according to an embodiment;

FIG. 10 is a view illustrating a process in which a circuit case is separated forward of a cabinet in a display apparatus shown in FIG. 9, according to an embodiment;

FIG. 11 is a rear view illustrating a display apparatus according to an embodiment;

FIG. 12 is an enlarged view of portion D of FIG. 11 when viewed at a different angle according to an embodiment; and

FIG. 13 is a view illustrating a coupling relationship between a first case and a second case in a display apparatus according to an embodiment.

MODES OF THE DISCLOSURE

Embodiments described in the specification and configurations shown in the accompanying drawings are merely examples of the disclosure, and various modifications may replace the embodiments and the drawings of the disclosure at the time of filing of the application.

Further, identical symbols or numbers in the drawings of the disclosure denote components or elements configured to perform substantially identical functions.

Further, terms used herein are only for the purpose of describing particular embodiments and are not intended to limit to the disclosure. The singular form is intended to include the plural form as well, unless the context clearly indicates otherwise. The terms “include,” “including,” “have,” and/or “having” specify the presence of stated features, integers, steps, operations, elements, components, and/or groups thereof, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

Further, although the terms “first,” “second,” etc. may be used herein to describe various elements, the elements are not restricted by the terms, and the terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and similarly, a second element could be termed a first element without departing from the scope of the disclosure. The term “and/or” includes combinations of one or all of a plurality of associated listed items.

The terms “front,” “rear,” “left,” “right”, and the like as herein used are defined with respect to the drawings, but the terms may not restrict the shape and position of the respective components.

A display apparatus according to an embodiment may include: a display module including a substrate on which

plurality of light emitting diodes (LEDs) are mounted; a cabinet provided to support the display module; and a circuit case accommodating electronic components therein, and coupled to the cabinet, wherein the circuit case is provided to be detachable from and mountable on to a front side of the cabinet at a front of the cabinet, and the circuit case is provided to be detachable from and mountable on a rear side of the cabinet at a rear of the cabinet.

The display apparatus may further include a connection bracket provided to couple the circuit case to the cabinet, wherein the connection bracket may be coupled to the cabinet at the rear of the cabinet by a fastening member.

The circuit case may be coupled to the connection bracket at the front of the connection bracket by a fastening member.

The circuit case may be separated forward of the cabinet by releasing the coupling between the connection bracket and the circuit case at the front of the cabinet.

The circuit case may be separated rearward of the cabinet by releasing the coupling between the connection bracket and the circuit case at the rear of the cabinet.

The circuit case may include: a first case having one side thereof open and accommodating electronic components therein; and a second case provided to cover the open side of the first case.

The first case may include a first hinge bracket including a shaft protrusion and coupled to one side surface of the first case.

The second case may include a second hinge bracket including a shaft hole into which the shaft protrusion may be inserted and coupled to one side surface of the second case.

The second case may be rotatably coupled to the first case by the shaft protrusion being inserted into the shaft hole, and the second case may be separated from the first case by the shaft protrusion being separated from the shaft hole.

The circuit case may further include a case sealing member provided along a side edge of the first case and sealing a region between the first case and the second case.

The display module may further include a reinforcing member coupled to a rear surface of the display module, the cabinet may include a plurality of first magnets, and the circuit case may further include a plurality of second magnets.

The display module may be coupled to the cabinet by magnetic attraction between the plurality of first magnets and the reinforcement member and magnetic attraction between the plurality of second magnets and the reinforcement member.

The display module may include a plurality of display modules, and the cabinet may be provided to support the plurality of display modules.

The cabinet may include a plurality of cabinets, and the plurality of cabinets may be provided to be coupled to each other in a horizontal direction and a vertical direction.

Hereinafter, embodiments according to the disclosure will be described in detail with reference to the accompanying drawings.

FIG. 1 is a view illustrating a display apparatus according to an embodiment. FIG. 2 is a view illustrating a cabinet and a display module separated from the cabinet in a display apparatus according to an embodiment, which is viewed from the front.

In FIG. 1, the X-axis, the Y-axis, and the Z-axis directions perpendicular to each other are shown, in which the X-axis direction refers to a left to right direction, the Y-axis direction refers to an upper to lower direction, and the Z-axis direction refers to a front to rear direction.

The following description is made with reference to FIGS. 1 and 2. A display apparatus 1 is a device that displays information, materials, data, and the like in the form of characters, figures, graphs, images, and the like, and may be implemented as a billboard, an electric sign board, a screen, a television, a monitor, and the like. The display apparatus 1 may be installed on a wall or ceiling, or installed on an indoor or outdoor ground by a stand.

The display apparatus 1 may include a display module 100 displaying a screen, a cabinet 200 coupled to a rear side of the display module 100 to support the display module 100, and a circuit case 300 coupled to the cabinet 200 and accommodating electronic components therein.

In the display apparatus 1 according to an embodiment, six display modules 100 may be coupled to one cabinet 200. Specifically, two of the display modules 100 may be coupled to the one cabinet 200 in a left to right direction X, and three of the display modules 100 may be coupled to the one cabinet 200 in an upper to lower direction Y. That is, the display modules 100 may be coupled to the cabinet 200 in a 2x3 matrix form. However, the number of display modules 100 coupled to the cabinet 200 is not limited. Two or more display modules 100 may be coupled to one cabinet 200.

Referring to FIG. 1, the display apparatus 1 according to the embodiment may include eight (8) cabinets 200 and forty eight (48) display modules 100 coupled to the eight cabinets 200. As described above, the display apparatus 1 may implement a large sized screen by consecutively arranging a plurality of cabinets 200, to which a plurality of display modules 100 are coupled, in the left to right direction X and the upper to lower upper to lower direction Y. The number of cabinets 200 consecutively arranged in the left to right direction X and in the upper to lower direction Y is not limited, and as described above, the number of the plurality of display modules 100 coupled to the cabinet 200 is not limited.

Referring to FIG. 2, the display module 100 may be separably coupled to the cabinet 200. The display module 100 may be separated from the cabinet 200 forward of the cabinet 200. The display module 100 may be coupled to the cabinet 200 by magnetic force. The cabinet 200 may include a plurality of first magnets provided on a front surface thereof. The circuit case 300 may include a plurality of second magnets provided on a front surface thereof. The display module 100 may include a reinforcement member (140, see FIG. 5) provided on a rear surface thereof and coupled to the plurality of first magnets and the plurality of second magnets by magnetic attraction. The plurality of first magnets provided on the cabinet 200 and the plurality of second magnets provided on the circuit case 300 may be the same components. The plurality of first magnets and the plurality of second magnets may be collectively referred to as the plurality of magnets 10.

The cabinet 200 may include frames (the first frame 210, the second frame 220, the third frame 230, and the fourth frame 240, see FIG. 6) forming an outer edge of the cabinet 200. The circuit case 300 may be disposed inside of the frames (the first frame 210, the second frame 220, the third frame 230, and the fourth frame 240). The circuit case 300 may include a second connector 301 corresponding to a first connector (112, see FIG. 5) included in each of the plurality of display modules 100. The circuit case 300 may include a first case 310 having a hexahedral shape with one surface open and a second case 320 provided to cover the open surface of the first case 310.

As the first connectors 112 of the plurality of display modules 100 are coupled to the second connectors 301 of the

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circuit case **300**, the plurality of display modules **100** may be electrically connected to each other. In addition, the plurality of display modules **100** may be supplied with power from a power supply device disposed inside the circuit case **300**.

FIG. **3** is an enlarged view of portion A of FIG. **2**.

Referring to FIG. **3**, the display module **100** may include a substrate (**110**, see FIG. **11**) and a holder **120** covering side and rear surfaces of the substrate **110** and supporting the substrate **110**. The substrate **110** may be attached to a front surface of the holder **120** through an adhesive, a double-sided adhesive tape, or the like, or coupled to the holder **120** through a fastening member, such as a screw.

A plurality of LEDs **111** may be mounted on the front surface of the substrate **110**. The substrate **110** may be formed of a material, such as glass, polyimide (PI), or FR4.

The plurality of LEDs **111** may include a red LED, a green LED, and a blue LED as one pixel in a package. A plurality of LEDs **111** may be mounted on the substrate **110** in a matrix form. The plurality of LEDs **111** may be arranged to have regular intervals, and intervals between the plurality of LEDs **111** may be variously determined according to the resolution and size of the display apparatus **1**.

The display module **100** may include a cover plate **130** provided to cover the front surface of the substrate **110**. The cover plate **130** may include a plurality of LED holes **131** respectively corresponding to the plurality of LEDs **111** mounted on the substrate **110**. The plurality of LEDs **111** may be exposed on the front surface of the cover plate **130** through the plurality of LED holes **131**.

The substrate **110** may be accommodated inside of the cover plate **130** and the holder **120**. The cover plate **130** and the holder **120** may be coupled to each other to form an inner space accommodating the substrate **110** therein.

The cover plate **130** may include a louver portion **132**. The louver portion **132** may include a horizontal louver portion **133** and a vertical louver portion **134**. The louver portion **132** may reduce or block moisture flowing into the plurality of LEDs **111**. In addition, the cover plate **130** may include a loop portion **135** provided on upper sides of the plurality of LEDs **111**. The loop portion **135** may protrude forward of the cover plate **130** to prevent water from being directly introduced into the plurality of LEDs **111** from above.

The cover plate **130** may include a fastening hole **136** into which a fastening member for coupling the cover plate **130**, the substrate **110**, and the holder **120** is inserted, and a fixing hole **137** into which a fixing member **20** is inserted.

FIG. **4** is a view illustrating a cabinet and a display module separated from the cabinet in a display apparatus according to an embodiment, which is viewed from the rear. FIG. **5** is a rear view illustrating a display module in a display apparatus according to an embodiment.

Referring to FIG. **4**, the display module **100** may be separated forward of the cabinet **200**. As described above, the display module **100** may be coupled to the cabinet **200** by magnetic force. A plurality of magnets (**10**, see FIG. **2**) may be provided on the front surface of the cabinet **200**, and a reinforcement member (**140**, see FIG. **5**) attracted by magnetic attraction of the plurality of magnets (**10**) may be provided on the rear surface of the display module **100**.

Referring to FIG. **5**, the reinforcement member **140** may be coupled to the rear surface of the holder **120**. The reinforcement member **140** may be disposed on an edge of the rear surface of the holder **120**. The reinforcement member **140** may include a first reinforcement member **141**, a second reinforcement member **142**, a third reinforcement member **143**, and a fourth reinforcement member **144**. The

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first reinforcement member **141** may be provided adjacent to an upper edge of the rear surface of the holder **120** and may extend in the horizontal direction. The second reinforcement member **142** may be provided adjacent to a lower edge of the rear surface of the holder **120** and may extend in the horizontal direction. The third reinforcement member **143** may be provided adjacent to a left edge of the rear surface of the holder **120** and may extend in the vertical direction. The fourth reinforcement member **144** may be provided adjacent to a right edge of the rear surface of the holder **120** and may extend in the vertical direction.

The reinforcement member **140** may be coupled to the rear surface of the holder **120** to reinforce the strength of the display module **100**. The reinforcement member **140** may prevent the display module **100** from bending. The reinforcement member **140** may prevent physical deformation of the display module **100**.

The reinforcement member **140** may be formed of a material that is attracted by the magnetic force of the magnet **10**. The reinforcement member **140** may be formed of a material that is magnetizable by an external magnetic force. The reinforcement member **140** and the magnet **10** may be attracted to each other by polarity of the reinforcement member **140** obtained from the external magnetic force. The reinforcement member **140** may include a magnetic material that interacts with the magnet **10**. The reinforcement member **140** may include iron (Fe).

The display module **100** may include a fixing member **20**. The fixing members **20** may be respectively provided on the upper side and the lower side of the rear surface of the display module **100**. According to one embodiment, the fixing members **20** may be provided at the upper center region of the rear surface and the lower center region of the rear surface of the display module **100**, but the positions thereof are not limited.

The fixing member **20** may prevent the display module **100** from being separated from the cabinet **200**. The display module **100** may be coupled by a magnetic attraction between the plurality of magnets **10** and the reinforcement member **140**, but in response to application of a force greater than the magnetic attraction to separate the display module **100** from the cabinet **200**, the display module **100** may be separated from the cabinet **200**. In order to prevent the separation, the fixing member **20** may be provided to prevent the display module **100** from moving forward of the cabinet **200**.

The display module **100** may include a first connector hole **128** that allows the first connector **112** provided on the rear surface of the substrate **110** to be exposed on the rear surface of the display module **100**. The display module **100** may include a first waterproof rib **122**, a second waterproof rib **123**, and a third waterproof rib **124** protruding rearward from the rear surface of the holder **120** to block water from being introduced into the first connector hole **128**. A sealing groove (**125**, see FIG. **7**) may be formed between the second waterproof rib **123** and the third waterproof rib **124**, and a sealing member **150** may be provided in the sealing groove **125**.

The first waterproof rib **122**, the second waterproof rib **123**, and the third waterproof rib **124** may each be provided in a ring shape. The second waterproof rib **123** may be provided inside of the first waterproof rib **122**, and the third waterproof rib **124** may be provided inside of the second waterproof rib **123**.

The first waterproof rib **122**, the second waterproof rib **123**, and the third waterproof rib **124** may be provided to be in contact with the front surface of the circuit case **300** to

prevent moisture from being introduced from the outer side to the inside of the first waterproof rib **122**, the second waterproof rib **123**, and the third waterproof rib **124**.

The sealing member **150** may be provided in the sealing groove **125** formed between the second waterproof rib **123** and the third waterproof rib **124**. The sealing member **150** may be provided to correspond to the shape of the sealing groove **125**. The sealing member **150** may be provided in a substantially ring shape. The sealing member **150** may be formed of a material capable of changing in volume, and may include, for example, rubber.

According to an embodiment, the sealing member **150** may be attached to the rear surface of the display module **100**, but is not limited thereto. The sealing member **150** may be attached to the front surface of the circuit case **300** to correspond to the sealing groove **125**.

The display module **100** may include a first waterproof rib **122a**, a second waterproof rib **123a**, a third waterproof rib **124a**, and a sealing groove **125a** bilaterally symmetrical to the first waterproof rib **122**, the second waterproof rib **123**, the third waterproof rib **124**, and the sealing groove **125**. The first waterproof rib **122a**, the second waterproof rib **123a**, and the third waterproof rib **124a** may be referred to as 'left side waterproof ribs,' and the first waterproof rib **122**, the second waterproof rib **123**, and the third waterproof rib **124** may be referred to as 'right side waterproof ribs.'

The left side waterproof ribs (the first waterproof rib **122a**, the second waterproof rib **123a**, and the third waterproof rib **124a**) may have the same structures bilaterally symmetrical to those of the right side waterproof ribs (the first waterproof rib **122**, the second waterproof rib **123**, and the third waterproof rib **124**). This is to allow the display modules **100** to be coupled to both a left column and a right column between two columns of the cabinet **200**. For example, when the display module **100** is coupled to the right column of the cabinet **200**, the first connector **112** may be disposed inside of the right side waterproof ribs (the first waterproof rib **122**, the second waterproof rib **123**, and the third waterproof rib **124**), and the sealing member **150** may be provided in the (right side) sealing groove **125** to prevent water from being introduced into the first connector **112**. Conversely, when the display module **100** is coupled to the left column of the cabinet **200**, the first connector **112** may be disposed inside of the left side waterproof ribs (the first waterproof rib **122a**, the second waterproof rib **123a**, and the third waterproof rib **124a**), and the sealing member **150** may be provided in the (left side) sealing groove **125a**. In addition, the first connector hole may not be provided inside of a waterproof rib in which the first connector **112** is not disposed. Referring to FIG. 5, a first connector hole may not be provided inside of the left side third waterproof rib **124a**, and a first connector hole **128** may be provided inside of the right side third waterproof rib **124** to allow the first connector **112** to be exposed there through.

In the following description, the right side waterproof ribs (the first waterproof rib **122**, the second waterproof rib **123**, and the third waterproof rib **124**) and the (right side) sealing groove **125** will be referred to as waterproof ribs (the first waterproof rib **122**, the second waterproof rib **123**, and the third waterproof rib **124**) and a sealing groove **125**.

FIG. 6 is a front view illustrating a cabinet in a display apparatus according to an embodiment.

The cabinet **200** may be formed by coupling (connecting, attaching) four frames (the first frame **210**, the second frame **220**, the third frame **230**, and the fourth frame **240**) each extending in one direction. The four frames (the first frame **210**, the second frame **220**, the third frame **230**, and the

fourth frame **240**) may be coupled to each other by four corner brackets (the first corner bracket **250a**, the second corner bracket **250b**, the third corner bracket **250c**, and the fourth corner bracket **250d**).

Specifically, the frames may include a first frame **210** forming an upper edge of the cabinet **200**, a second frame **220** forming a lower edge of the cabinet **200**, a third frame **230** forming a left edge of the cabinet **200**, and a fourth frame **240** forming a right edge of the cabinet **200**. The first frame **210** and the second frame **220** may be disposed in the horizontal direction, and the third frame **230** and the fourth frame **240** may be disposed in the vertical direction.

The first frame **210** and the fourth frame **240** may be coupled to each other by a first corner bracket **250a**. The first frame **210** and the third frame **230** may be coupled to each other by a second corner bracket **250b**. The second frame **220** and the third frame **230** may be coupled to each other by a third corner bracket **250c**, and the second frame **220** and the fourth frame **240** may be coupled to each other by a fourth corner bracket **250d**.

The first frame **210**, the second frame **220**, the third frame **230**, and the fourth frame **240** may be coupled to each other by the first corner bracket **250a**, the second corner bracket **250b**, the third corner bracket **250c**, and the fourth corner bracket **250d** to have a rectangular frame shape.

The circuit case **300** may be located at the central region inside of the frames (the first frame **210**, the second frame **220**, the third frame **230**, and the fourth frame **240**). The circuit case **300** may be coupled to the first frame **210** and the second frame **220** by a first connection bracket **340** and a second connection bracket **350** respectively provided on the upper side and the lower side of the circuit case **300**.

The circuit case **300** may include a first case **310** forming front and side surfaces of the circuit case **300** and having a rear surface thereof open, and a second case **320** provided to cover the open rear surface of the first case **310**. The first case **310** and the second case **320** may be coupled to each other to form the circuit case **300**. The first case **310** and the second case **320** may be coupled to each other so that an inside of the first case **310** and the second case **320** may be sealed. The first case **310** and the second case **320** may be coupled to each other to block inflow of water from the outside.

The first case **310** may have a second connector hole **311a** formed through the front surface thereof. The first case **310** may include a rib **311** provided along the circumference of the second connector hole **311a** and protruding forward. A second connector **301** may be disposed inside of the second connector hole **311a**.

The second connector hole **311a**, the second connector **301**, and the rib **311** may be provided corresponding in number to the number of display modules **100** coupled to the cabinet **200**.

Inside the circuit case **300**, provided may be a power supply device provided to supply power to the display module **100**, a control board provided to control the display module **100**, and the like. In this specification, electronic parts may include the power supply device and the control board described above.

The circuit case **300** may include a support bracket **330** provided to support the display module **100**. The support brackets **330** may be coupled to both sides of the circuit case **300**. In addition, the support brackets **330** may be provided above and below each other, on both sides of the circuit case **300**. According to an embodiment, four support brackets **330** may be coupled to the circuit case **300**.

The cabinet **200** may include a plurality of magnets **10**. In addition, the circuit case **300** may include a plurality of magnets **10**.

Specifically, the cabinet **200** may include a plurality of magnets **10** disposed spaced apart from each other on the front surfaces of the first frame **210**, the second frame **220**, the third frame **230**, and the fourth frame **240**.

In addition, the circuit case **300** may include a plurality of magnets **10** disposed spaced apart from each other on the front surface of the circuit case **300**.

Referring to FIG. 6, the cabinet **200** and the circuit case **300** according to the embodiment may include ten magnets **10** for each of the display module **100**. The cabinet **200** and the circuit case **300** may include a total of sixty magnets **10**. The cabinet **200** may include thirty eight (38) magnets **10**, and the circuit case **300** may include twenty two (22) magnets **10**. However, the number and position of the magnets **10** are not limited thereto, and the number and position of the magnets **10** may be changed.

For example, in order to support the display module **100** coupled to the upper right side of the cabinet **200**, the cabinet **200** may include the first to seventh magnets **10a**, **10b**, **10c**, **10d**, **10e**, **10f**, and **10g**, and the circuit case **300** may include the eighth to tenth magnets **10h**, **10i**, and **10j**.

As described above, the plurality of magnets **10** may attract the reinforcement member **140**, which is formed of a magnetizable material and disposed on the rear surface of the display module **100**, by magnetic attraction, so that the display module **100** may be coupled to the cabinet **200**.

FIG. 7 is an enlarged view of portion B of FIG. 4.

According to an embodiment, the fixing member **20** may be provided in the form of a wing nut. Specifically, the fixing member **20** may include a body portion **21** having a cylindrical shape and inserted into a holder hole **121** of the holder **120** and a protrusion portion **22** protruding outward of the body portion **21**. The protrusion portions **22** may be provided as a pair.

The cabinet **200** may include a through hole **201** through which the fixing member **20** may pass. The fixing member **20** may be provided to be rotatable between a disengagement position in which the fixing member **20** is allowed to pass through the through hole **201** and an engagement position in which the fixing member **20** is prevented from passing through the through hole **201** by the protrusion portion **22** being caught in the through hole **201**.

The user may rotate the fixing member **20**, which protrudes rearward of the cabinet **200** through the through hole **201**, at the rear of the cabinet **200** by hands without a separate tool, thereby engaging the display module **100** with the cabinet **200**, or disengaging the display module **100** from the cabinet **200**. When the fixing member **20** passed through the through hole **201** is in the disengagement position and a force greater than or equal to magnetic force is applied to the display module **100**, the display module **100** may be separated from the cabinet **200**. When the fixing member **20** passed through the through hole **201** is in the engagement position, the display module **100** may be prevented from being separated from the cabinet **200**.

The display module **100** may include a fixing hole (**137**, see FIG. 3) through which at least a portion of the fixing member **20** is exposed on the front surface of the display module **100**. The fixing hole **137** may be formed by passing through the display module **100**. One end of the fixing member **20** may be exposed on the front surface of the display module **100** through the fixing hole **137**. The one end of the fixing member **20** may be provided with a groove having a predetermined shape. For example, the one end of

the fixing member **20** may be provided with the same hexagonal groove **23** as that provided at the other end of the fixing member **20**.

A user may rotate the fixing member **20** at the front of the display module **100** using a separate tool having a shape corresponding to the groove formed in the one end of the fixing member **20**. As described above, the display module **100** may become separable from the cabinet **200** by rotating the fixing member **20**. Through this, the display module **100** may be separated from the cabinet **200** even at the front of the display module **100** without approaching the rear surface of the cabinet **200**.

As described above, the display module **100** may be coupled to the cabinet **200** by the magnetic attraction between the magnet **10** and the reinforcement member **140**. In order to prevent the display module **100** from being separated from the cabinet **200** in response to application of an external force greater than or equal to the magnetic attraction, the display module **100** may include the fixing member **20**. When the fixing member **20** is positioned at the engagement position, the display module **100** may be prevented from being separated from the cabinet **200**.

FIG. 8 is a view illustrating a cabinet and a circuit case separated from the cabinet in a display apparatus according to an exemplary embodiment.

Referring to FIG. 8, the circuit case **300** may be detachably coupled to the cabinet **200**. The circuit case **300** may be coupled to the cabinet **200** by the first connection bracket **340** and the second connection bracket **350**.

The first connection bracket **340** may couple the upper side of the circuit case **300** to the first frame **210**. The second connection bracket **350** may couple the lower side of the circuit case **300** to the second frame **220**.

As will be described below, the circuit case **300** may be separated from the cabinet **200** at the front of the cabinet **200**, and the circuit case **300** may be separated from the cabinet **200** at the rear of the cabinet **200**.

The circuit case **300** may further include a coupling bracket **360** coupling the first case **310** and the second case **320** together. As the coupling bracket **360** is coupled to the first case **310** and the second case **320**, the first case **310** and the second case **320** may be coupled to each other.

FIG. 9 is an enlarged view of portion C of FIG. 6 when viewed at a different angle. FIG. 10 is a view illustrating a process in which a circuit case is separated forward of a cabinet in a display apparatus shown in FIG. 9.

As described above, the fixing member **20** may be rotated even at the front of the display apparatus **1** using a separate tool. After rotating the fixing member **20** to a disengaged position, the plurality of display modules **100** may be detached from the cabinet **200**.

In a state in which the plurality of display modules **100** are separated from the cabinet **200**, the front surface of the circuit case **300** may be exposed to the front of the display apparatus **1** as shown in FIG. 9.

In order to separate the circuit case **300** forward of the display apparatus **1**, a first fastening member **30** exposed on the front side of the first connection bracket **340** may be removed. The first fastening member **30** may be provided in various forms. For example, the first fastening member **30** may include a screw.

In FIG. 10, as the first fastening member **30** exposed on the front side of the first connection bracket **340** is removed, the circuit case **300** may be separable from the first connection bracket **340**. The separated circuit case **300** may be slightly moved rearward and downward of the cabinet **200** and then withdrawn forward of the cabinet **200**.

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The first connection bracket **340** may include (i) a first fastening hole **341** into which the first fastening member **30** is inserted and/or fastened and (ii) a first guide hole **342** into which a first guide protrusion **361** of the coupling bracket **360** is inserted. As the first guide protrusion **361** is inserted into the first guide hole **342**, the location of the circuit case **300** may be guided before the circuit case **300** is coupled to the first connection bracket **340** and the second connection bracket **350**.

FIG. **11** is a rear view illustrating a display apparatus according to an embodiment. FIG. **12** is an enlarged view of portion D of FIG. **11** when viewed at a different angle.

Referring to FIG. **11**, the rear surface of the circuit case **300** may be exposed on the rear side of the cabinet **200** without the display module **100** being separated from the cabinet **200**. When there is a need to separate the circuit case **300** rearward of the cabinet **200**, the first connection bracket **340** and the second connection bracket **350** may be separated from the cabinet **200**. As the first connection bracket **340** and the second connection bracket **350** are separated from the cabinet **200**, the circuit case **300**, the first connection bracket **340**, and the second connection bracket **350** may be separated from the cabinet as one part.

The first connection bracket **340** may include a second guide protrusion **343** provided to guide a coupling position of the first connection bracket **340** with respect to the cabinet **200**. The first frame **210** of the cabinet **200** may include a second guide hole **211** into which the second guide protrusion **343** is inserted. The first frame **210** may include a second fastening hole **212** spaced laterally from the second guide hole **211** and into which the first fastening member **30** is inserted.

The first connection bracket **340** may include a third fastening hole **344** into which the first fastening member **30** is inserted. The second fastening hole **212** and the third fastening hole **344** may be provided to correspond to each other.

A second fastening member **40** may couple the second case **320** to the first case **310**. The second fastening member **40** may be coupled to the coupling bracket **360** by passing through the second case **320**, to thereby coupling the second case **320** to the first case **310**.

FIG. **13** is a view illustrating a coupling relationship between a first case and a second case in a display apparatus according to an embodiment.

Referring to FIG. **13**, a case sealing member **380** may be provided between the first case **310** and the second case **320**. The case sealing member **380** may be provided along the circumference of the first case **310**. The case sealing member **380** may be provided to seal a region between the first case **310** and the second case **320**.

The first case **310** may include a first hinge bracket **370**. The first hinge bracket **370** may include a shaft protrusion **371**.

The second case **320** may include a second hinge bracket **321**. The second hinge bracket **321** may include a shaft hole **322**.

The shaft protrusion **371** may be provided to be rotatable within the shaft hole **322**. The second case **320** may be separated from the first case **310** only by lifting the second case **320** upward while the shaft protrusion **371** is inserted into the shaft hole **322**.

Conversely, by simply positioning the second case **320** on the first case **310** such that the shaft protrusion **371** is inserted into the shaft hole **322**, the second case **320** may be

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rotatably coupled to the first case **310**. In addition, the second case **320** may open and close the open rear surface of the first case **310**.

Although the disclosure has been shown and described in relation to specific embodiments, it would be appreciated by those skilled in the art that changes and modifications may be made in these embodiments without departing from the principles and scope of the disclosure, the scope of which is defined in the claims and their equivalents.

What is claimed is:

1. A display apparatus comprising:
 - a display module comprising a substrate on which a plurality of light emitting diodes (LEDs) are mounted;
 - a cabinet provided to support the display module; and
 - a circuit case attached to the cabinet, wherein the circuit case is provided to be detachable from and mountable on a front side of the cabinet, wherein the circuit case is provided to be detachable from and mountable on a rear side of the cabinet, wherein the circuit case comprises: a first case comprising an open side; and a second case provided to cover the open side of the first case, and wherein the first case accommodates electronic components,
 - wherein the display module further comprises a reinforcing member attached to a rear surface of the display module, wherein the cabinet comprises a plurality of first magnets, and wherein the circuit case further comprises a plurality of second magnets; and
 - wherein the display module is attached to the cabinet by a first magnetic attraction between the plurality of first magnets and the reinforcing member, and wherein the display module is attached to the cabinet by a second magnetic attraction between the plurality of second magnets and the reinforcing member.
2. The display apparatus of claim 1, further comprising a connection bracket provided to attach the circuit case to the cabinet, wherein the connection bracket is attached to the rear side of the cabinet by a fastening member.
3. The display apparatus of claim 2, wherein the circuit case is attached to the connection bracket at a front side of the connection bracket by the fastening member.
4. The display apparatus of claim 3, wherein the circuit case is separable from the front side of the cabinet by detaching the connection bracket from the circuit case at the front side of the cabinet.
5. The display apparatus of claim 3, wherein the circuit case is separable from the rear side of the cabinet by detaching the connection bracket from the cabinet at the rear side of the cabinet.
6. The display apparatus of claim 1, wherein the first case further comprises a first hinge bracket comprising a shaft protrusion, and wherein the first hinge bracket is attached to a first side surface of the first case.
7. The display apparatus of claim 6, wherein the second case comprises a second hinge bracket comprising a shaft hole, wherein the shaft protrusion is inserted into the shaft hole, and wherein the second hinge bracket is attached to a second side surface of the second case.
8. The display apparatus of claim 7, wherein the second case is rotatably connected to the first case.
9. The display apparatus of claim 6, wherein the second case comprises a second hinge bracket comprising a shaft hole,

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wherein the second hinge bracket is attached to a side surface of the second case, and wherein the second case is separated from the first case by the shaft protrusion that is separated from the shaft hole.

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10. The display apparatus of claim **1**, wherein the circuit case further comprises a case sealing member provided along a side edge of the first case, and

wherein the case sealing member is provided to seal a region between the first case and the second case.

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11. The display apparatus of claim **1**, further comprising a plurality of display modules, and wherein the cabinet is provided to support the plurality of display modules.

12. The display apparatus of claim **1**, wherein the cabinet is included in a plurality of cabinets, and wherein the plurality of cabinets are provided to be connected to each other in a horizontal direction and a vertical direction.

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